

Towards MAPPO and Inverse DRL for MATSim

Replanning

A dimensionless person-level learning framework for scalable transport simulation

Extended Abstract for the MATSim User Meeting

Abstract. This project explores how multi-agent proximal policy optimization (MAPPO), an extension of proximal policy optimization (PPO), can be integrated into MATSim at the person level. The implementation replaces the conventional replanning portfolio with a shared actor-critic policy that proposes mode, rerouting, and departure-time actions for each traveller while retaining MATSim’s QSim and scoring machinery as the execution and behavioural-evaluation environment. A central design goal is to avoid network-specific input dimensions: states are expressed through normalized, dimensionless route, time, congestion, and person features rather than fixed link-by-link vectors. The extension supports fixed inference, scenario-specific retraining, and custom network architectures, together with imitation learning, optional on-policy updates, count-aware routing, and experienced transit timetable updates. The design is also informed by inverse deep reinforcement learning (inverse DRL): calibrated MATSim utility scores and demonstrated default replanning decisions provide observable behavioural signals for policy learning.

Keywords: MATSim; MAPPO; PPO; inverse DRL; reinforcement learning; replanning; agent-based transport simulation; QSim; transit; counts.

1. Motivation: PPO and Inverse DRL

MATSim represents daily travel demand through individual persons, plans, and repeated execution-replanning cycles. Its established replanning strategies, including route choice, trip-mode choice, and time allocation mutation, are transparent and robust. At the same time, their effects are selected from a configured strategy portfolio and evaluated over many iterations. This project investigates a complementary approach: treating each person as a homogeneous deep reinforcement-learning agent whose policy can learn a coordinated replanning rule from MATSim’s own simulated outcomes. The purpose is not to replace QSim or MATSim scoring. Rather, it is to replace the decision logic that generates alternative plans while preserving the established behavioural and traffic-flow simulation core.

The work is motivated by the recent success of PPO-style policy-gradient reinforcement learning in large multi-agent decision problems. Here, a single parameter-sharing MAPPO policy serves the population, while a critic evaluates each local decision against a richer system context. The architecture is intended as an experimental path toward more adaptive, reusable, and computationally structured replanning for large-scale transport scenarios. The project also connects to inverse DRL: observed travel diaries or calibrated behavioural choices can be treated as evidence of latent preferences, while MATSim’s score function supplies a structured utility signal. The current implementation does not yet infer a reward function directly from trajectories; instead, it uses imitation learning and score-based rewards as a practical first step toward that inverse-learning formulation.

2. Framework Design

Figure 1 summarises the implementation. QSim remains the execution engine and produces the event stream, experienced travel conditions, and plan outcomes. During scoring, the extension collects state-action-reward sequences (SARS) for valid activity-to-activity trips, excluding stage and interaction activities. The actor selects a masked multi-head action comprising a travel mode, a rerouting choice, and a discrete departure-time shift. The critic estimates the value of the corresponding system state. MAPPO-generated plans are marked as selected for the next QSim run, so they are executed and scored in exactly the same way as ordinary MATSim plans.

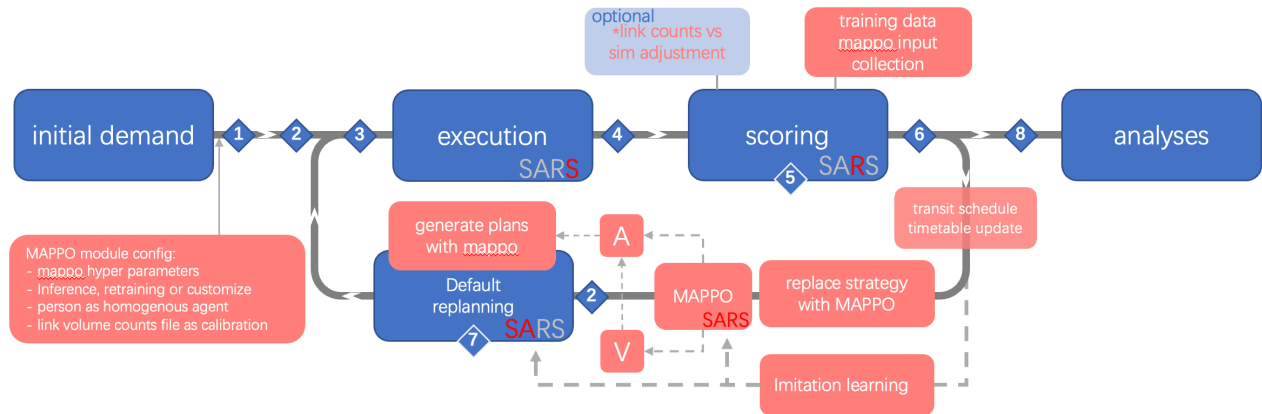


Figure 1: MAPPO integration with the MATSim demand, replanning, QSim execution, scoring, and analysis loop.

3. Dimensionless State Representation and Actions

A practical obstacle to transferring learned policies between cities is that a raw network state has a different number of links, nodes, and public-transport elements in every scenario. The proposed representation therefore does not feed a fixed network tensor directly to the policy. Instead, it uses dimensionless or capped features describing the current decision point: normalized time of day, leg ordinal, previous mode, licence and vehicle availability, beeline and route distance, route travel time relative to free flow, link-count and V/C summaries along the route, and optional count-versus-simulation discrepancies. The actor receives only decision-relevant local features; the critic additionally receives aggregate congestion and observed-count context. This creates a fixed input dimension even when network size changes, while still allowing QSim to supply scenario-specific traffic conditions.

Actions are feasibility-masked before softmax. The mode head follows the configured MATSim modes and can remove infeasible alternatives according to scenario settings, person attributes, subtour continuity, and distance rules. The rerouting and departure-time heads are likewise constrained by routing and time-allocation feasibility. The result is a policy output that can be decoded directly into a valid MATSim plan rather than an abstract action requiring a separate behavioural interpreter.

4. Operating Modes and Learning Lifecycle

The MAPPO config group exposes three user-facing operating modes. In inference mode, a user can load pretrained actor and critic checkpoints to replace legacy strategies such as ReRoute, ChangeTripMode, and TimeAllocationMutator. With `onPolicy=false`, the loaded policy remains fixed and performs direct plan prediction. With `onPolicy=true`, the same loaded policy continues to learn from completed MAPPO

rollouts after each iteration. This supports a controlled distinction between a reproducible deployment run and an adaptive online experiment.

Retraining is intended for a scenario whose scoring parameters, mode parameters, or demand assumptions have been calibrated and should be reflected in a policy of the standard architecture. Customized mode reserves the same workflow for a user-defined hidden-layer configuration or other compatible policy design. In both cases, ordinary MATSim replanning operates until a configurable takeover iteration. The resulting default strategy decisions form an imitation-learning dataset. MAPPO then takes over plan generation. After takeover, `onPolicy=true` enables PPO updates from MAPPO’s own state-action trajectories; `onPolicy=false` freezes the imitation-trained policy and uses it only for inference.

5. Count and Transit Enhancements

Two additional test-scenario enhancements connect the policy loop to observed and experienced operations. First, when a MATSim counts file is available, the extension exposes normalized simulated-versus-observed link-volume differences to the critic and can apply a count-aware impedance adjustment to car rerouting. The objective is not to force every individual decision to match a count, but to make the routing environment more responsive where systematic count discrepancies are observed. Second, the extension records transit vehicle arrival and departure events in QSim and uses previous-iteration experienced stop offsets to update the next iteration’s transit schedule. This makes PT routing reflect simulated vehicle operations rather than an unchanged nominal timetable.

6. Preliminary Test Evidence and Next Steps

Initial 10,000-person test scenarios confirm the end-to-end feasibility of the integration: MAPPO plans can be generated, executed, scored, retained or rejected against a prior selected plan, and written as rollout data for offline inspection. The tests also demonstrate that the dimensionless state schema, action masks, count-aware features, and experienced transit schedule updates can operate inside the normal controller loop. In preliminary takeover tests, person plan scores improved relative to the immediate pre-takeover baseline while policy outputs, mode shares, route statistics, total travel time, and count-fit metrics remained available through standard MATSim outputs. These results should be read as integration evidence rather than a claim of general superiority: fair cross-city transfer and convergence tests remain necessary.

The next development steps are to improve cross-scenario validation, reduce online-training cost, test policy transfer across networks with different topology and scale, and analyse Pareto trade-offs between individual utility, system travel time, modal outcomes, and count fit. The same architecture can subsequently be extended beyond persons. Potential agents include transit vehicles for service and timetable optimization, signal controllers for network control, and other operational actors. In this sense, the project is an attempt to connect state-of-the-art multi-agent learning with the behavioural detail and policy relevance of MATSim-based transport planning.